Calc 1 — Assignment 5

1. (3 pts)

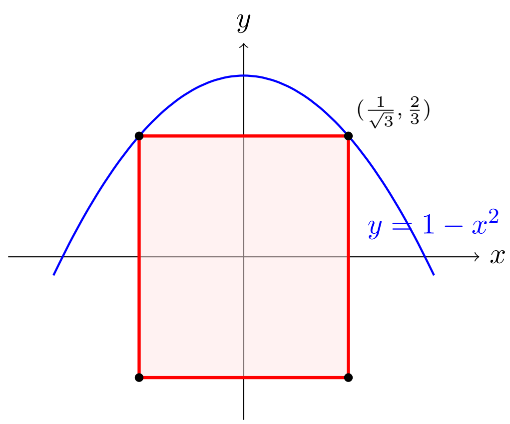
The rectangle has width $2a$, height $2b$, and $b = 1 - a^2$ with $0 \leq a \leq 1$. So

$$A(a) = 4ab = 4a(1 - a^2).$$

Set $A'(a) = 4 - 12a^2 = 0$, giving $a = \frac{1}{\sqrt{3}}$ (and $A'' < 0$, so it's a max). Then $b = \frac{2}{3}$ and

$$\boxed{A_{\text{max}} = \frac{8\sqrt{3}}{9}.}$$

The sides are $2a = \frac{2\sqrt{3}}{3} \approx 1.155$ and $2b = \frac{4}{3} \approx 1.333$, so it's **not a square** — it's a bit taller than wide.



2. (3 pts)

Falling from rest, $v^2 = 2gh$:

$$v = \sqrt{2(9.8)(400)} = \sqrt{7840} = 28\sqrt{10} \approx 88.54 \text{ m/s}.$$

That's under 90, so **yes, it survives**.

3. (2 pts)

(i)

Differentiating $\int_{-x}^{x} f = 0$ gives $f(x) + f(-x) = 0$, so f has to be odd. Two examples: $f(t) = t$ and $f(t) = \sin t$.

(ii)

Same calculation as (i): the condition is exactly $f(x)+f(-x) = 0$. So the answer is **any continuous odd function**.

(iii)

Trick question — there's no such function (other than $f \equiv 0$).

Let $G(x) = \int_{-x}^{2x} f$. If $G \equiv 0$, then

$$G'(x) = 2f(2x) + f(-x) = 0.$$

Plug in $x = 0$: $3f(0) = 0$, so $f(0) = 0$. Swapping $x \to -x$ gives $f(x) = -2f(-2x)$, and using the same relation on $f(-2x)$ gives $f(-2x) = -2f(4x)$. Combining,

$$f(x) = 4f(4x), \quad \text{equivalently} \quad f\left(\tfrac{x}{4^n}\right) = 4^n f(x).$$

As $n \to \infty$, the left side $\to f(0) = 0$ by continuity, so $4^n f(x) \to 0$, which forces $f(x) = 0$. So no nonconstant continuous example exists.

4. (4 pts)

(a)

Differentiating both sides of $\dfrac{\pi}{2} + \int_a^x f = x \sin x$ gives

$$f(x) = \frac{d}{dx}(x \sin x) = \sin x + x \cos x.$$

Plugging $x = a$ into the original equation kills the integral, leaving $\frac{\pi}{2} = a \sin a$. Try $a = \frac{\pi}{2}$: $\frac{\pi}{2} \sin \frac{\pi}{2} = \frac{\pi}{2}$. ✓

$$\boxed{f(x) = \sin x + x \cos x, \qquad a = \tfrac{\pi}{2}.}$$

(b)

Chain rule + FTC:

$$f'(x) = \frac{x^2 - 3}{2 - \sin(3x^2)} \cdot 2x = \frac{2x(x - \sqrt{3})(x + \sqrt{3})}{2 - \sin(3x^2)}.$$

The denominator is always ≥ 1, so the sign of f' matches the numerator. Sign chart:

	$x < -\sqrt{3}$	$-\sqrt{3} < x < 0$	$0 < x < \sqrt{3}$	$x > \sqrt{3}$
f'	$-$	$+$	$-$	$+$

So:

$$\boxed{x = -\sqrt{3}: \text{ min}, \quad x = 0: \text{ max}, \quad x = \sqrt{3}: \text{ min.}}$$

5. (3 pts)

(a)

Differentiate $xy = 1$: $y + xy' = 0$, so $y' = -y/x = -1/a^2$ at $(a, 1/a)$. Perpendicular slope is the negative reciprocal:

$$\boxed{m_\perp = a^2.}$$

(b)

Write $P = (x, 1/x)$ and minimize the squared distance to $Q = (0, 2)$:

$$D^2(x) = x^2 + \left(\tfrac{1}{x} - 2\right)^2.$$

At the closest point, $\dfrac{d}{dx}D^2 = 0$:

$$2x + 2\left(\tfrac{1}{x} - 2\right)\left(-\tfrac{1}{x^2}\right) = 0 \implies x^4 + 2x - 1 = 0. \qquad (\star)$$

Now the slope of QP is $m_{QP} = \dfrac{1/x - 2}{x} = \dfrac{1 - 2x}{x^2}$, and from (a) the tangent slope is $-1/x^2$. So QP is perpendicular to the curve iff

$$m_{QP} \cdot \left(-\tfrac{1}{x^2}\right) = -1 \iff \frac{1 - 2x}{x^2} = x^2 \iff x^4 + 2x - 1 = 0,$$

which is exactly $(\star)$. So at the closest point, $QP \perp$ curve. $\square$